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Posterior cortical atrophy misdiagnosed as typical Alzheimer's disease: Clinical and neuropsychological evolution over eight years. A case report

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Abstract

Posterior cortical atrophy (PCA) is a challenging diagnosis and is often misdiagnosed due to its subtle and atypical presentation. We present a woman with early-onset visuospatial dysfunction, apraxia, and dysgraphia. Neuroimaging revealed posterior atrophy with sparing of the hippocampi. Clinical features, neuropsychological findings, and neuroimaging were consistent with a diagnosis of PCA. The diagnosis was established based on the three-level PCA classification framework. This case highlights the importance of clinical evaluation and multidisciplinary collaboration in identifying atypical presentations of neurodegenerative diseases.

Keywords

Alzheimer's disease, cognitive impairment, dementia, posterior cortical atrophy

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Introduction

Posterior cortical atrophy (PCA) is a clinico-radiological syndrome characterized by neurodegeneration of the parietal and occipital lobes.[1] It manifests clinically with early visuoperceptual impairment and symptoms suggestive of Gerstmann and Balint syndromes.[2] Although episodic memory is typically preserved in the early stages, disease progression ultimately results in a widespread pattern of cognitive impairment.[3,4] Patients with PCA present with an atypical and non-amnesic cognitive decline, which often leads to a delayed or misdiagnosis.[5]

The most common underlying cause of PCA is Alzheimer's disease (AD), with a recent meta-analysis reporting that almost 94% of PCA cases have neuropathological evidence of AD.[2] However, PCA may also result from other neurodegenerative processes, including Lewy body disease (PCA-LBD), corticobasal degeneration (PCA-CBD), and prion disease (PCA-Prion).[6] In most cases, diagnosis is made clinically using the three-level classification proposed by Crutch et al. Access to biomarkers to determine the underlying etiology is limited. Consequently, clinicians will have to rely on comprehensive clinical evaluations and collaborative efforts to achieve diagnostic clarity.

We present a 68-year-old woman with PCA, exhibiting worsening visuospatial abilities, parietooccipital atrophy, bilateral myoclonus and late-stage visual hallucinations, who was initially misdiagnosed with typical AD. We discuss the clinical reasoning process, differential diagnosis, and challenges faced in a low-resource setting, pointing out the necessity of increasing access to advanced diagnostic methods while efficiently using clinical and cognitive assessments.

Case presentation

We report the case of a woman who presented to our center with a diagnosis of AD. The onset of symptoms began

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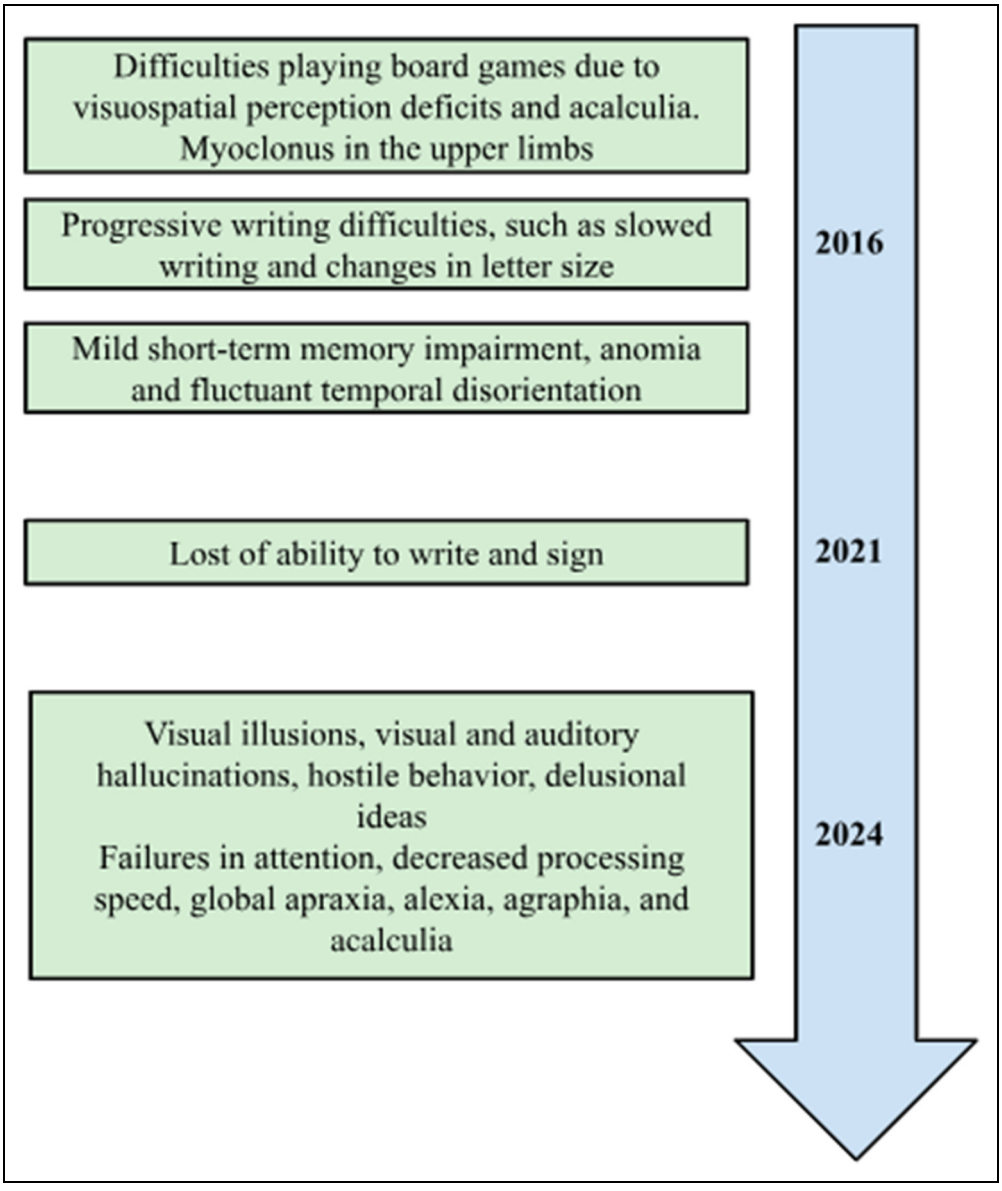
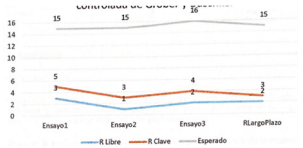
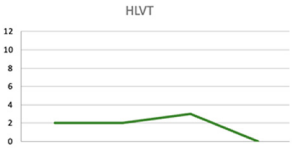
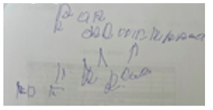
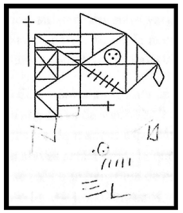
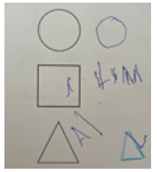
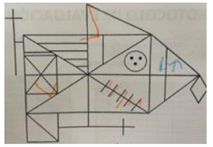


Figure 1. Timeline of symptom progression.

around 2016, when she was 60 years old. She experienced problems with board games due to a visuospatial perception deficit, where she would make mistakes like moving backwards instead of forward, struggle to place game pieces correctly and encounter difficulties with counting, alongside experiencing unexplained hand movements described as myoclonus. Subsequently, she developed difficulties with writing ("she wrote one large letter, then a small one"), which progressed to a complete loss of writing ability. In the following years, additional symptoms emerged: difficulty dressing and right/left disorientation. Her family also noted that she struggled to reach and grasp objects, even when they were clearly visible. Less prominent symptoms consisted of mild retrograde amnesia and anomia (Figure 1).

Due to these symptoms, she consulted neurology, underwent neuropsychological assessment (Table 1) and magnetic resonance imaging (MRI) (Figures 2 and 3), and was diagnosed with AD. Treatment was initiated with rivastigmine patches (9.5 mg), escitalopram (20 mg/day), and levetiracetam (1000 mg every 12 h). In 2024, she was brought to our clinic by her relatives, who expressed concerns about the diagnosis. In 2024, the patient began experiencing behavioral symptoms, including visual illusions and hallucinations, auditory hallucinations, hostile behavior, and delusional ideas. Our assessment revealed oculomotor apraxia, hyperreflexia, polymyoclonus in both hands, a positive palmomental reflex, finger agnosia, left/right disorientation, and both dressing and ideomotor apraxia. A third comprehensive neuropsychological assessment

Table 1. Evolution of symptoms in neuropsychological tests.

DOMAIN	2020	2022	2024
Attention and orientation	Preserved in time, space, and person. Displays impaired attention and concentration, associated with reduced graphomotor skills. Temporal, spatial, and personal orientation remains intact	Attention deficits center on short and prolonged periods. Diminished selective attention and alternating attention between stimuli and between various tasks. Preserved in space and person.	Diminished focused attention, and visual sustained and selective attention. Auditory digit retention compromised. Low processing of speed. Preserved in space and person.
Memory	Learning curve is slightly ascending, with no further improvement past the second trial, despite repeated presentations. Loss of recently learned information. **Memory curve:** 	Reduced, downward, and unproductive learning curve, showing a severely limited learning capacity and an inability to use or benefit from facilitation strategies for information encoding. There was a clear loss of information over time. **Grober and Buschke test:** 	Deficits in encoding, storage, recall and retrieval verbal information. Poor performance in visual episodic and semantic memory tests. **Hopkins Verbal Learning Test:** 
Language	Altered denomination and sentence repetition. Preserved comprehension of commands, word repetition, reading, writing, comprehension of concepts, as well as expressive and receptive communication skills.	Low performance in verbal fluidity (worst in semantic areas), and a high quantity of pathological phenomena such as circumlocutions and semantic and phonological paraphasias. Loss of writing abilities such as writing a letter.	Language expression and comprehension below expected performance, phonological paraphasias and multiple anomias. Possible verbal apraxia. Alexia, agraphia and acalculia.  Visuoconstructional apraxia shown in the Rey complex figure. Impairment in the ability to discriminate right and left. Dressing apraxia, apraxia of speech, ocular apraxia and difficulties focusing on all visual tasks.
Praxis and Visuospatial process	Mild impairment in the execution of visuospatial tasks such as cube copying, interlaced pentagons, infinity symbol, and clock test. Visuoperceptual and visuoconstructional apraxia. NO IMAGES AVAILABLE FROM THE FIRST ASSESSMENT	Persistent alterations in visuospatial, visuoperceptual, and visuoconstructional abilities according to performance on the Rey Figure test. The TMT A amd B and symbol-digit-modalities test were discontinued. 	 
Executive functions	Alterations in the abstraction, planning, sequencing, inhibition, attentional control and flexibility capacities. Impairment working memory.	Very low performance in tasks related to working memory, diminished calculation skills, and low abstraction and categorization capacities. Deficits in cognitive flexibility.	Significant low performance on tasks requiring organization, calculation, planning, working memory, inhibitory control, and verbal fluency tasks.

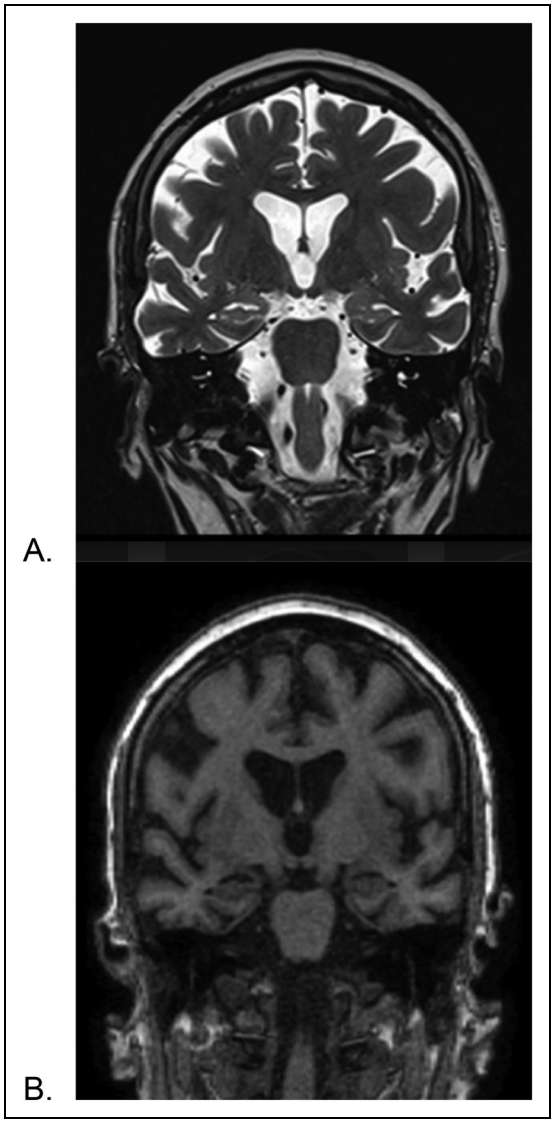


Figure 2. Cortical atrophy. (A) Coronal MRI, T2: 2022. (B) Coronal MRI, T1: 2024.

was carried out, encompassing appraisals of both functional and behavioral domains (Tables 1 and 2). Screening assessments revealed significant impairments in multiple areas, encompassing difficulties with motor sequencing, instruction processing, motor and verbal inhibitory control, auditory and visual working memory, and abstraction.

Behavioral questionnaires indicated concerns about self-neglect, disorganization, inattention, and verbal apraxia. The Cummings Neuropsychiatric Inventory Questionnaire (NPI-Q) and the Frontal Behavioral Inventory were administered, revealing delusions, hallucinations, agitation, depression, anxiety, apathy, irritability, and disturbances in sleep and nighttime behavior (Table 2).

Overall, the neuropsychological findings revealed a substantial decrease in both cognitive and functional abilities. The ability to distinguish left from right was severely impaired with significant difficulties in visual construction tasks. In contrast, personal and spatial orientation remained preserved (Table 1). Language testing showed reduced processing speed and difficulties in both expressive and receptive language, along with anomia and multiple phonemic paraphasias. Impaired word articulation suggested the presence of verbal apraxia. Reading, writing, and calculation were also impaired, consistent with alexia, agraphia, and acalculia. Memory assessment showed significant impairments in encoding, storing, and recalling both visual and verbal information.

Further workup, including MRI (Figures 2 and 3) and laboratory testing, was performed to investigate secondary or reversible causes of the behavioral symptoms and hallucinations. Results of complete blood count, vitamin B12, HIV-Elisa, VDRL, folic acid, glycated hemoglobin, and renal, hepatic, and thyroid function were all within normal limits, which rules out metabolic or infectious causes.

This case describes a woman with chronic and progressive major cognitive impairment of early onset (beginning around age 60). The initial manifestations included impairments in visual-perceptual skills, apraxia, myoclonus, agraphia, and Gerstmann syndrome, with relatively less memory involvement. In advanced stages psychotic and behavioral symptoms have emerged. Neuroimaging revealed progressive lobar atrophy (Figure 3), predominantly in posterior regions, with minimal to no involvement of the hippocampi (Figure 2). The clinical picture is suggestive of PCA syndrome.

Rivastigmine dosage was increased to 27 mg/day, and quetiapine was started at a dose of 12.5 mg every 12 h, in conjunction with the implementation of non-pharmacological strategies for managing hallucinations and behavioral symptoms. However, the patient continued to experience hallucinations, prompting a gradual increase in quetiapine to 100 mg/day, without clinical improvement. Consequently, levetiracetam was discontinued, and valproic acid along with risperidone 2 mg/day were introduced. Following the onset of extrapyramidal symptoms after this change, valproic acid was substituted with lacosamide. Currently, the patient has satisfactory control of psychotic symptoms, with no significant changes in cognitive functioning.

Discussion

This case describes a woman with a progressive major neurocognitive disorder of early onset. Initial symptoms included visuospatial disturbances, apraxia, dysgraphia, and myoclonus, along with predominantly posterior brain atrophy. These findings are suggestive of PCA syndrome. In most cases, PCA is caused by abnormal accumulation of amyloid-β (Aβ) and tau proteins in posterior brain

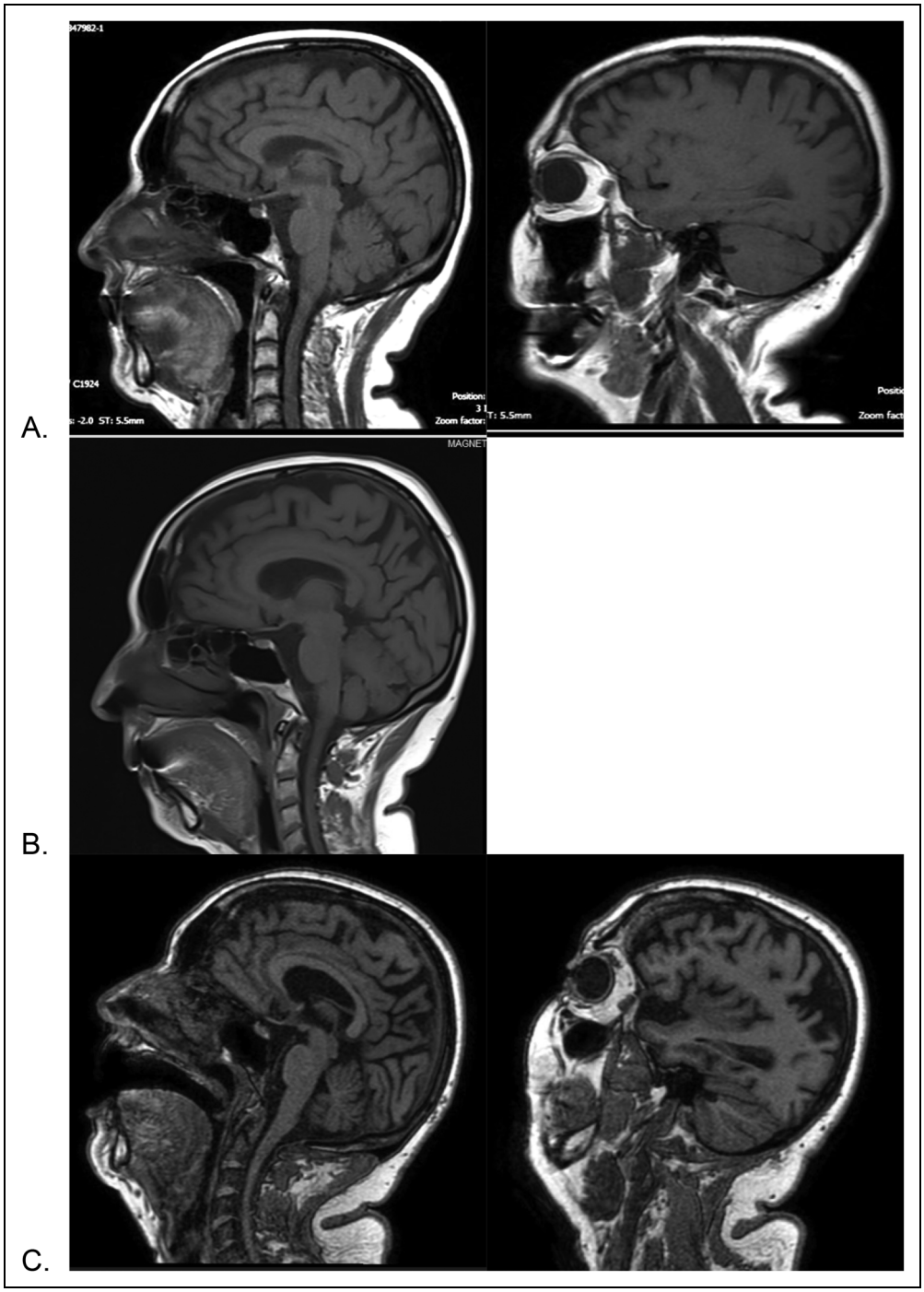


Figure 3. Sagittal and parasagittal MRI, T1 shows progressive atrophy in parietal and occipital cortex through years. (A) 2018. (B) 2022. (C) 2024.

regions, making it one of the atypical clinical presentations of AD.[7]

This clinico-radiological syndrome can also be caused by other neurodegenerative diseases, such as dementia with Lewy bodies (DLB) or corticobasal degeneration (CBD).[3] In our case, we did not observe typical symptoms of DLB, including fluctuating cognition, early visual hallucinations, spontaneous parkinsonism or REM sleep behavior disorder. Although this patient presented neuroleptic sensitivity and hallucinations appeared, they did so at an advanced stage of the disease and no other core characteristics of DLB were identified.[8] CBD typically presents with symptoms such as asymmetric limb rigidity or akinesia, dystonia, and/or myoclonus, as well as limb apraxia,

Table 2. Screening tests and questionnaire results 2024 neuropsychological assessment.

Dimension	Test		Score		Analysis
			Expected	Patient	
Global performance	INECO Frontal Screening	Motor series	> 25/30	0/3	Inferior
		Conflicting instructions		0/3	
		Motor inhibitory control		0/3	
		Digit span backwards		2/6	
		Months backward		0/2	
		Visual working memory		0/4	
		Proverbs		0/3	
		Inhibitory control		0/6	
		TOTAL		2/30	
Functionality	Activities of Daily Living	Barthel Index	PD 100	PD 70	Dependent
	Instrumental Activities of Daily Living	Lawton and Brody scale	PD 8	1-0-0-1-0-1-0-0	Dependent
Behavior	Frontal Behavioral Inventory		PD < 24	PD 40	Significant changes
	Cummings Neuropsychiatric Inventory Questionnaire (NPI-Q)		–	Delusions YES	S3 ; F4
				Hallucinations YES	S3 ; F4
				Agitation NO	–
				Depression YES	S2 ; F2
				Anxiety NO	–
				Joy NO	–
				Apathy YES	S2 ; F3
				Disinhibition YES	S2 ; F2
				Irritability YES	S2 ; F2
				Motor disturbance YES	S1 ; F1
				Sleep disturbance YES	S1 ; F 2
				Appetite NO	–

S: Severity; F: Frequency.

cortical sensory deficits, and alien limb phenomena[3]; in this case, the patient only exhibited myoclonus without displaying the other core characteristics.

Since the onset of symptoms, the progression observed in neuropsychological assessments performed over the past years has revealed an initial decline in visuospatial and visuoconstructional domains. This deterioration has also impacted attentional and executive functioning. A mild decline in episodic memory was identified, particularly in retrieval processes; however, this domain remained relatively preserved compared to the visuospatial impairments.

Although the clinical picture initially raised suspicion of possible AD, it is well established that typical presentations of this neurodegenerative condition usually begin with prominent memory complaints, while visuospatial and visuoconstructional abilities tend to remain intact in the early stages—contrary to what was observed in our patient.

In the latest neuropsychological evaluation, all cognitive domains showed significant impairment. The patient demonstrated low processing speed, anosognosia, anomia, verbal apraxia, alexia, agraphia, acalculia, and marked dysexecutive symptoms. There was also a notable disturbance in left-right discrimination and profound visuoconstructional deficits. Importantly, although episodic memory was affected, no intrusions or false positives were observed—an uncommon finding in the typical course of AD, further supporting the atypical nature of this case.

The diagnosis of PCA was based on the Crutch et al. 2017 framework, which proposes a three-level classification system. Level 1 refers to the presence of clinical and radiological features consistent with PCA. Level 2 distinguishes between a "pure" PCA presentation and "PCA-plus," where features of other neurodegenerative syndromes coexist, particularly when biomarker data are unavailable. Level 3 aims to identify the underlying pathology through the use of biological markers. According to this framework, the patient meets level 1 criteria: insidious onset, gradual progression, prominent early disturbance of visual cognitive functions, left/ right disorientation, dressing apraxia, agraphia, constructional dyspraxia, relative spared memory at onset and parieto-occipital atrophy. Regarding level 2 criteria non core characteristics of DLB nor CBS were identified. Level 3 could not be established, as access to biomarkers and molecular diagnosis are limited. Biological confirmation is primarily used to guide inclusion in clinical trials or to initiate disease-modifying therapies, but it is not essential for symptomatic treatment.[9] Therefore, understanding the clinical presentation, symptom progression, and neuroimaging features of the various neurodegenerative disorders that cause neurocognitive impairment is essential—particularly in settings where biological confirmation is limited.[10] Identifying PCA as a unique condition is essential for patients and their families, enabling tailored pharmacological and non-pharmacological treatment approaches that address visuoperceptual impairment and posterior cortical neurodegeneration.

Conclusion

Establishing a proper diagnosis helps the clinician to rule out non-neurodegenerative causes and guide pharmacological and neuropsychological management. Moreover, for patients and their families, a differential diagnosis can elucidate the clinical course, prognosis, and potential heritability risk. Even in the absence of curative treatments, accurately distinguishing AD from other dementia subtypes remains crucial, as it ensures individualized therapeutic approaches, enables realistic prediction of disease course and fosters more effective communication with patients and caregivers regarding disease expectations and long-term planning.

Although significant progress has been made with biological and molecular tools for diagnosing dementia, these resources are not yet widely accessible. Consequently, clinical criteria remain the primary basis for diagnosis in certain areas. Healthcare professionals must remain open to the full spectrum of cognitive disorders, rather than assuming all cognitive decline is due to typical AD. Taking into account the timeline of symptom onset, the age at presentation, and the pattern of cognitive changes can lead to more accurate clinical classification. Although PCA is rare, it is a late or unrecognized diagnosis.

Finally, decisions regarding biological diagnosis of the underlying pathologic processes should be individualized, considering patient age, disease stage, atypical or rapidly progressive features, eligibility for clinical trials or disease-modifying therapies, and, most importantly, whether the results would alter management or prognosis.

This case highlights the significance of collaborative work between neurologists and neuropsychologists in making a precise diagnosis and in handling and tracking patients with PCA.

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Ethical considerations

This study was conducted in accordance with the principles of the Declaration of Helsinki.

Consent to participate

Written informed consent was obtained from the patient prior to participation.

Consent for publication

Written informed consent for publication of their clinical details and/or clinical images was obtained from the relative of the patient. A copy of the consent form is available for review by the editor of this journal. Patient confidentiality was safeguarded by anonymizing all personal information and clinical images.

Author contribution(s)

Mateo Avilés-Gómez: Investigation; Methodology; Writing – original draft; Writing – review & editing.
Suhamny Navas-Peñaranda: Writing – original draft.
Maria P Hurtado-Barreto: Conceptualization; Investigation; Writing – original draft; Writing – review & editing.
Mariana Gaviria-Carrillo: Conceptualization; Formal analysis; Investigation; Project administration; Visualization; Writing – original draft; Writing – review & editing.

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Declaration of conflicting interests

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